MONASH CLAYTON SCHOOL OF MEDICINE
2025 Year 1 Semester 1 Lecture Curriculum

Semester at a Glance

Yr 1 Semester 1 Lecture Timetable, 2025

Theme	Week 1 (3-Mar)	Week 2 (12-Mar)	Week 3 (19-Mar)	Week 4 (24-Mar)	Week 5 (31-Mar)	Week 6 (7-Apr)	Week 7 (14-Apr)	Week 8 (28-Apr)	Week 9 (5-May)	Week 10 (12-May)	Week 11 (19-May)	Week 12 (26-May)
Personal & Professional Development			**Ethics and biomedical ethics** ETHICS - Introduction to ethics (PD)	**Ethics and biomedical ethics** ETHICS - Introduction to biomedical ethics (PD)	**Doctor/patient relationships** ETHICS - Medical Paternalism (PD)	**Doctor/patient relationships** ETHICS - Informed consent and confidentiality (PD)						
				Integrative Medicine /Behavioural change HEP - Introduction to Integrative Medicine (CHas)	**Stress** HEP - Introduction - stress and mind-body interactions (CHas)			**Exercise** HEP - Nutrition (CHas)		**Connectedness/Spirituality & environment** HEP - Environment (CHas)	**Connectedness/Spirituality & environment** HEP - Connectedness (CHas)	
				HEP - Behaviour change, lifestyle and motivation (CHas)	HEP - Mindfulness (CHas)			HEP - Exercise (CHas)			HEP - Spirituality (CHas)	
Population, Society & Illness	**Knowledge Structures** HKS - Health and medicine (AP)	**Health Systems** HKS - The national health system (AP)	**Complex Inequity** HKS - Health Determinants (AP)		**Complex Inequity** HKS - Indigenous Health (AP)		**Complex Inequity** HKS - Primary health care in practice (AP)	**Complex Inequity** HKS - Access to healthcare (AP)		**Creating Health Justice** HKS - Refugee & asylum seeker health (AP)	**Creating Health Justice** HKS -Health & human rights (AP)	**Creating Health Justice** HKS - The pharmaceutical industry (AP)
Foundations of Medicine	**Cell Basics** ANAT - Cells I (CH)	**Cell Basics** BIOCHEM - Molecular Biology of the Cell 1 - Transcription & Translation (NH)	**Cell Basics** BIOCH - Integrated metabolism (NH)	**Cell Basics** BIOCH - Understanding the relevance of genomics to medicine (MP/NH)	**Cell to Cell Communication** ANAT - Connective Tissue (JB)	**Cell to Cell Communication** ANAT - Bones & Cartliage (JB)	**Cell to Cell Communication** ANAT - Histology of Muscle Tissue (JB)	**Cell to Cell Communication** PATH - Blood composition & haemopoiesis (TBC)	**Cellular enemies** PATH -Cellular response to injury Part I-III (TBC)	**Cellular Enemies** IMMUN - Innate response to pathogens (KM)	**Cellular Enemies** MICRO - Antimicrobial resistance & resistance transfer (MH)	
	ANAT - Cells II (CH)	BIOCHEM - Molecular Biology of the Cell 2 - Genetic replication (NH)	BIOCH - Population genetics (MP/NH)	Cell to Cell Communicaiton ANAT - Epithelium (JB)		PHYS - Resting membrane potential & action potential (SC)	PHARM - Introduction to pharmacology (BKH)	PATH - Haemostasis & coagulation (TBC)	MICRO - Introduction to microbes (MH)	IMMUN - The adaptive immune system (KM)	MICRO - Parasites (MH)	
	ANAT - Cell cycle (JY)	BIOCHEM - Cellular energetics: proteins and enzymes (NH)	BIOCH - Chromosome disorders (NH)			PHYS - Action potential & propagating the action potential (SC)	PHARM - Drugs & neurotransmission (BKH)	**When Cells make Mistakes** PHARM - Anticancer drugs (BKH)	MICRO - Natural barriers (MH)	IMMUN - Infection & Immunity (KM)	PHARM - Intro to pharmacokinetics (BKH)	
	PHYS - Body fluids (BS)	BIOCH - Fuel molecules and energetics (NH)	BIOCH - Applications of gene technologies to medicine (NH)			PHYS - synaptic transmission (SC)	When Cells make Mistakes BIOCH - Introduction to cancer (NH)		MICRO - Strategies & conrtol (MH)		PHARM - Antibiotics (JBour)	
						PHYS - Spinal cord, neural communication & spinal reflexes (SC)	BIOCH - What effect does cancer have on the patient (NH)		MICRO - Fungal infections (MH)		PHARM - Antivirals (JBour)	
				HD -Theories of human development (IL)	HD - Adolescent development (IL)					HD - Normal Ageing (IL)		HD - Cognitive Development (IL)
				HD - Infancy and early childhood development (IL)	HD - Adult development (IL)					HD - Death and Dying (IL)		HD - Introduction to health and human behaviour (IL)

Lecturer Key		Lecturer Key		Discipline Key
				ANAT - Antaomy
AP	Dr Asika Pelenda	JY	Dr Julia Young	BIOCH - Biochemistry
BKH	A/Prof Barb Kemp-Harper	KM	Dr Kim Murphy	ETHICS - Ethics
BS	Dr Ben Seyer	MH	A/Prof Meredith Hughes	HD - Human Development
CH	Dr Chantal Hoppe	MP	Dr Maude Phipps (Malaysia)	HEP - Health Enhancement Program
CHas	Prof Craig Hassed	NH	Dr Nathan Habila	HKS - Health, Knowledge & Society
IL	Dr Izaak Lim	PD	Dr Peter Douglas	IMMUN - Immunology
JB	Prof John Bertram	SC	Dr Simone Carron	MICRO - Microbiology
JBour	Prof Jane Bourke			PHYS - Physiology
				PHARM - Pharmacology

Week 1

Lecture 1

Theme: Population, Society & Illness

Session Outline: HKS - Health and medicine

Learning Outcomes
• Understand health and its complexities • Examine intersectionality / biosocial interactions that influence health outcomes • Appraise dominant thought structures within medical discourse • Distinguish between the role of structure, culture, history, and demographic influences on health • Evaluate sources of global health inequity using a social justice lens

Lecture 2

Theme: Foundations of Medicine

Session Outline: ANAT - Cells I

Learning Outcomes
• Describe the general features of eukaryotic cells; nucleus components and function • Describe the basic features and function of cell organelles including mitochondria, ribosomes, RER, SER, Golgi, lysosomes, secretory vesicles • Describe the structural components of the cell membrane (carbs, lipids, proteins) and their function; features of biological membranes • Describe the major components and function of the cytoskeleton including microtubules, actin, myosin, intermediate filaments • Describe the composition and function of some examples (centrioles, cilia, flagella, microvilli) • Identify cells & their organelles using light microscopy, electron microscopy

Lecture 3

Theme: Foundations of Medicine

Session Outline: ANAT - Cells II

Learning Outcomes
• Describe the general features of eukaryotic cells; nucleus components and function • Describe the basic features and function of cell organelles including mitochondria, ribosomes, RER, SER, Golgi, lysosomes, secretory vesicles • Describe the structural components of the cell membrane (carbs, lipids, proteins) and their function; features of biological membranes • Describe the major components and function of the cytoskeleton including microtubules, actin, myosin, intermediate filaments. • Describe the composition and function of some examples (centrioles, cilia, flagella, microvilli) • Identify cells & their organelles using light microscopy, electron microscopy

Lecture 4

Theme: Foundations of Medicine

Session Outline: ANAT – Cell cycle

Learning Outcomes
• Understand the sequence of the mitotic cell cycle • Know the major drivers of the process – cyclin/CDK interactions • Understand how the cycle can be stopped - regulation of the cell cycle • Describe how the cycle is different when making gametes • Understand the difference in meiosis during spermatogenesis and oogenesis • Understand the complications in meiosis • Understand the two major ways cells can die – apoptosis and necrosis • Understand why cells die during development and in disease states

Lecture 5

Theme: Foundations of Medicine

Session Outline: PHYS – Body fluids

Learning Outcomes
• Describe how different molecules cross the plasma membrane • Explain the effects of varying tonicity on cells in the body • Give a basic account how various clinical conditions can affect fluid and electrolyte balance

Week 2

Lecture 1

Theme: Population, Society & Illness

Session Outline: HKS - The National Health System

Learning Outcomes
• Describe the key components of the Australian health system • Explain the sources of health funding of the Australian health system • Identify key components of the Australian health system • Discuss the importance of ‘systems thinking’ in health systems • Give examples to illustrate the role of adverse patient outcomes in the improvement of healthcare delivery • Analyze complex relationships within the national health systems

Lecture 2

Theme: Foundations of Medicine

Session Outline: BIOCH - Molecular biology of the cell 1 – Transcription and Translation

Learning Outcomes
• Describe the structure of DNA and its organisation in the cell • Explain the role of RNA in expressing genetic information • Explain the process of protein synthesis

Lecture 3

Theme: Foundations of Medicine

Session Outline: BIOCH - Molecular biology of the cell 2 – Genetic replication

Learning Outcomes
• Describe the structure of DNA and its organisation in the cell • Explain the steps in the replication of DNA • Explain the need for a robust DNA repair system

Lecture 4

Theme: Foundations of Medicine

Session Outline: BIOCH - Cellular energetics: proteins and enzymes

Learning Outcomes
• Describe the structure and levels of protein organisation • Explain the function of proteins in biological systems • Describe enzymes and their role in biochemical reactions • Explain the factors which govern biological reactions

Lecture 5

Theme: Foundations of Medicine

Session Outline: BIOCH - Fuel molecules and energetics

Learning Outcomes
• Outline the process of catabolism of glucose, lipids and proteins ▪ Discuss how energy production is linked to the availability of oxygen ▪ Compare energy production from different fuel molecules (carbohydrates and lipids) ▪ Explain in outline what happens when the body has an excess of energy - how does the body make use of this energy?

Week 3

Lecture 1

Theme: Personal and Professional Development

Session Outline: ETHICS - Introduction to ethics

Learning Outcomes
• Have a deeper understanding of the nature of ethics • Be familiar with the basic tenets of four important moral frameworks in biomedical ethics: consequentialism, deontology, communitarianism, and virtue ethics

Lecture 2

Theme: Population, Society & Illness

Session Outline: HKS - Health determinants

Learning Outcomes
• Explain the influence of key social factors on health and illness • Outline the wide range of factors interacting to influence health status • Describe the concepts of health inequality and inequity • Explain the links between the social determinants of health and health inequities • Explain the effects of the social determinants of health on clinical and public health practice

Lecture 3

Theme: Foundations of Medicine

Session Outline: BIOCH - Integrated metabolism

Learning Outcomes
• Outline the specialised metabolic pathways occurring in the liver, adipose tissue, skeletal and heart muscle, brain and red blood cells • Discuss the interrelationship of carbohydrate, lipid and protein metabolism in the above tissues • Integrate metabolism in the whole organism

Lecture 4

Theme: Foundations of Medicine

Session Outline: BIOCH - Population genetics

Learning Outcomes
• Discuss fundamental aspects of population genetics/genomics and diversity • Describe genetic structure by allele, genotype frequencies, Hardy-Weinberg equilibrium • Describe diversity of traits and how genetic disorders may be more common in one population than another

Lecture 5

Theme: Foundations of Medicine

Session Outline: BIOCH - Chromosome disorders

Learning Outcomes
• Define what a karyotype is • Describe the nomenclature used for chromosome analysis • Identify and explain the different classes of chromosome abnormality • Explain how chromosomal abnormalities occur and their impact on health and diseases

Lecture 6

Theme: Foundations of Medicine

Session Outline: BIOCH - Applications of gene technologies to medicine

Learning Outcomes
Part 1 Recombinant DNA Technology & Therapeutics: • Discuss the basic procedures for creating and cloning recombinant DNA • Explain the basic principles of polymerase chain reaction (PCR) • Explain how DNA technology can be used to produce large quantities of therapeutics
Part 2 Therapeutic Cloning: • Define the concept of therapeutic cloning • Describe the different approaches to therapeutic cloning
Part 3 Gene Therapy & Gene Editing: • Explain the basic principle of gene therapy • Describe how CRISPR-Cas9 gene editing works • Explain how gene editing can be used in gene therapy

Week 4

Lecture 1

Theme: Personal and Professional Development

Session Outline: ETHICS - Introduction to biomedical ethics

Learning Outcomes
• Explain the role of the four principles of respect for autonomy, non-maleficence, beneficence, and distributive justice in shaping ethical reasoning in healthcare • Appreciate the basic differences between ethics and the law

Lecture 2

Theme: Personal and Professional Development

Session Outline: HEP - Introduction to integrative medicine

Learning Outcomes
• Discuss terminology relating to Integrative Medicine (IM) • Discuss relevance of IM to modern healthcare • Discuss community attitudes to and usage of complementary and alternative medicine (CAM) • Discuss uptake of CAM among doctors in Australia • Discuss main modalities of CAM • Discuss main concerns with regard to CAM use • Discuss lessons gleaned from the CAM phenomenon • Discuss a few key ethical and clinical issues relating to CAM • Discuss where to access further information and training on CAM

Lecture 3

Theme: Personal and Professional Development

Session Outline: HEP - Behaviour change lifestyle and motivation

Learning Outcomes
• Understand the factors associated with better health and self-care among medical students • Appreciate the importance of lifestyle factors in promoting longevity and the development and progression of chronic illness • Revise the basic elements of the ESSENCE model • Describe the role of education in healthcare (BASK) • Understand the importance of enabling strategies such as: – ADEPT model of changing habits – Motivational interviewing – Goal setting, with particular reference to the SMART model – Prochaska-Di Clemente model of behaviour change • Integrate this knowledge with weekly cases – E.g. immunity, cancer and heart disease • Understand how lifestyle programs are integrated in healthcare

Lecture 4

Theme: Foundations of Medicine

Session Outline: BIOCH - Understanding the relevance of genomics to medicine

Learning Outcomes
• Appreciate how knowledge of genetics and genomics can inform the understanding of, and impact the practice of medicine • Understand how genomics/genetics informs disease characterization, diagnostics, molecular pathology, pharmacogenomics, gene therapy, biotechnology applications, etc. • Reflect upon some ethical issues that new developments in genetics and genomics may raise health in terms of health and wellbeing of patients, families and communities

Lecture 5

Theme: Foundations of Medicine

Session Outline: ANAT - Epithelium

Learning Outcomes
• Understand the definition, distribution, characteristics and functions of epithelia • Understand the different types of epithelia found throughout the body • Understand the structure and function of specialisations at the apical, lateral and basal surfaces of epithelia • Appreciate that both exocrine and endocrine glands are derived from epithelia

Lecture 6

Theme: Foundations of Medicine

Session Outline: HD – Theories of human development

Learning Outcomes
• Define development and ageing, and their relationship to each other • List and illustrate the seven key assumptions of the modern lifespan perspective • Outline the four issues addressed by theories of human development • Summarise the three parts of the personality and the five psychosexual stages in Freud's psychoanalytic theory • Explain the conflicts humans face according to Erikson’s eight psychosocial stages • Differentiate between classical conditioning, operant conditioning, and observational learning • Outline the main emphases of humanistic theories of development • Summarise the epigenetic psychobiological systems perspective of human development

Lecture 7

Theme: Foundations of Medicine

Session Outline: HD – Infancy & early childhood development

Learning Outcomes
• Name newborn reflexes and explain their significance • Describe the heritable and environmental influences on temperament • Describe and identify stages of physical and motor development from birth to 2 years of age • Classify biological, psychological and social influences on prenatal development

Week 5

Lecture 1

Theme: Personal and Professional Development

Session Outline: ETHICS - Medical paternalism

Learning Outcomes
• Appreciate the ethical dimensions of clinical relationships and the historical role of medical paternalism.

Lecture 2

Theme: Personal and Professional Development

Session Outline: HEP - Introduction - stress and mind-body interactions

Learning Outcomes
• Understand the role of mental health in the burden of disease • Understand the stress and relaxation responses • Appreciate the effects of stress on body, mental health and lifestyle • Understand the relationship between stress and performance • Appreciate the effects of perception and cognition on stress

Lecture 3

Theme: Personal and Professional Development

Session Outline: HEP - Mindfulness

Learning Outcomes
• Discuss the importance of attention and attentional training • Discuss principles of mindfulness-based stress management and cognitive therapy • Discuss clinical applications of mindfulness-based strategies

Lecture 4

Theme: Population, Society & Illness

Session Outline: HKS - Indigenous health

Learning Outcomes
• Explain the terms “Indigenous people” • Describe the unique health risks and determinants of health related to Indigenous populations • Explain the disparity in health status between the Indigenous population and Aboriginal and Torres Strait Islander People and the non-Indigenous Australian population • Outline key government initiatives related to Indigenous health in Australia • Explain the impact of social factors on the health of Indigenous Australians • Describe the barriers to attaining the Close the Gap Campaign goals • Discuss the progress of the targets (benchmarks) of the Closing the Gap strategy • Explain the impact of health professionals and general societal behaviour on access to health care for Indigenous people

Lecture 5

Theme: Foundations of Medicine

Session Outline: ANAT - Connective tissue

Learning Outcomes
• Know the different types of connective tissue • Understand the similarities and differences between the different types of connective tissue

Lecture 6

Theme: Foundations of Medicine

Session Outline: HD - Adolescent development

Learning Outcomes
• Describe physical development associated with puberty • Summarise the developmental tasks of adolescence • Describe health conditions prevalent in adolescents • Understand the main threats to health with a focus on obesity and risky behaviour and possible underlying causal factors • Identify biological, social and behavioural aspects of coupling and sexual behaviour in adolescence

Lecture 7

Theme: Foundations of Medicine

Session Outline: HD - Adult development

Learning Outcomes
• Demonstrate an understanding of adult social relationships including friendship, marriage and divorce • Demonstrate an appreciation of domestic violence and its impacts, at an introductory level • Demonstrate an understanding of the decision to have children and parenting roles and duties • Demonstrate a capacity to describe the importance of work and career and the need for work/life balance • Demonstrate an appreciation of the impact of unemployment • Demonstrate an understanding of the importance and role of leisure during adulthood and its impact on wellbeing

Week 6

Lecture 1

Theme: Personal and Professional Development

Session Outline: ETHICS - Informed consent and confidentiality

Learning Outcomes
• Understand what is meant by informed consent and appreciate its ethical implications • Understand the ethical obligations that arise from concerns with privacy and confidentiality in clinical contexts

Lecture 2

Theme: Foundations of Medicine

Session Outline: ANAT - Bones & cartilage

Learning Outcomes
• Understand the histology and functions of hyaline, elastic and fibro cartilage • Understand the histology and functions of compact and spongy bone

Lecture 3

Theme: Foundations of Medicine

Session Outline: PHYS - Resting membrane potential & action potential

Learning Outcomes
• Understand that neurons have a difference in electrical charge across their membrane • Describe the processes for electrical charge in the semi-permeable membrane and ion movement • Apply the Nernst and Goldman-Hodgkin-Katz equation for calculating resting membrane potentials • Learn why a semi-permeable membrane can create a difference in electrical charge across the membrane, and how this can be applied to the nerve cell membrane

Lecture 4

Theme: Foundations of Medicine

Session Outline: PHYS - Action potential & propagating the action potential

Learning Outcomes
• Understand the difference between a graded and an action potential • Understand the concept of threshold, and the properties of the ion channels involved • Understand the events that return the membrane potential to the resting level after the action potential • Understand the concept of refractory periods, the two refractory periods that occur after an AP and how action potentials are propagated • Understand the role of ion pumps/transporters in recovery

Lecture 5

Theme: Foundations of Medicine

Session Outline: PHYS - Synaptic transmission

Learning Outcomes
• Describe the anatomy of a chemical synapse/electrical synapse • Describe general concepts of neurotransmitter production • Know how neurotransmitters are released and inactivated at the synapse • Describe the major types of neurotransmission, and the most common excitatory and inhibitory neurotransmitters in the CNS • Understand the role of IPSP and EPSPs • Explain how the activation of receptors by neurotransmitters is translated into a response

Lecture 6

Theme: Foundations of Medicine

Session Outline: PHYS - Spinal cord, neural communication and spinal reflexes

Learning Outcomes
• Describe the organization and structure of the spinal cord and spinal nerves • Describe the main ascending and descending tracts • Compare and contrast between spinal reflexes - monosynaptic and polysynaptic reflexes

Week 7

Lecture 1

Theme: Population, Society & Illness

Session Outline: HKS - Primary healthcare in practice

Learning Outcomes
• Explain the concept of primary health care • Describe the conditions recommended to enable efficient primary health care • Develop an understanding of the use of health indicators to explain between-country differences in population health, which are used to measure the effectiveness of Primary health care and health systems • Discuss the concept of community development

Lecture 2

Theme: Foundations of Medicine

Session Outline: ANAT - Histology of muscle tissue

Learning Outcomes

• Understand the similarities and differences in the three types of muscle (skeletal, cardiac, smooth) in terms of their:
  ○ structure
  ○ function
  ○ distribution
  ○ innervation
• Be able to explain the concepts of the sarcomere and the sliding filament theory
• Describe the composition of layers of a blood or lymphatic vessel wall
 	
  • *tunica intima, tunica media, tunica adventitia, and elastic laminae*
 	
• Compare and contrast different types of blood and lymphatic vessels with regard to vessel function and histological differences in wall structure
 	
  • Artery (elastic and muscular), arteriole, capillary, venule, vein, lymphatic capillary
 	
• Identify different types of blood vessels using histological images

Lecture 3

Theme: Foundations of Medicine

Session Outline: PHARM - Introduction to pharmacology

Learning Outcomes
• Describe the field of 'pharmacology' • Describe the principles of drug nomenclature • List some useful sources of information about drugs

Lecture 4

Theme: Foundations of Medicine

Session Outline: PHARM - Drugs and neurotransmission

Learning Outcomes
• Describe the criteria that need to be satisfied for a chemical to be designated as a neurotransmitter • Describe the basic mechanisms of synthesis, storage, release and termination of action of neurotransmitters • Describe the chain of events which occur after an action potential arrives at the neuromuscular junction that eventually produces contracture of the skeletal muscle • Describe mechanisms by which neurotransmission can be modulated • Explain the actions of <u>non-depolarising drugs</u> at the skeletal neuromuscular junction • Explain the actions of <u>depolarising drugs</u> at the skeletal neuromuscular junction • Describe the strategies for treatment of myasthenia gravis

Lectures 5 and 6

Theme: Foundations of Medicine

Session Outline: BIOCH - Introduction to cancer, and what effect does cancer have on the patient

Learning Outcomes
• Recognise the characteristics and behaviour of cancer cells • Explain benign and malignant cancer • Explain the tissue effects of localised versus metastatic cancer • Explain the effects of accumulated multiple DNA errors/mutations • Define how mutations in tumour suppressors and oncogenes predispose to cancer development • Describe the features of inherited cancer susceptibility syndrome

Week 8

Lecture 1

Theme: Personal and Professional Development

Session Outline: HEP - Nutrition

Learning Outcomes
• Understand the health impacts of poor nutrition and the therapeutic potential for food as medicine • Name and describe the many psychological and social factors which impact upon eating patterns • Implement some of the simple behavioural and psychological strategies for improving eating patterns

Lecture 2

Theme: Personal and Professional Development

Session Outline: HEP - Exercise

Learning Outcomes
• Understand the many health benefits of regular exercise including the amount and intensity of exercise associated with those benefits • Recommend the appropriate type, amount and intensity of exercise to meet various health needs • Know what exercise prescriptions are and how they are used in healthcare settings

Lecture 3

Theme: Population, Society & Illness

Session Outline: HKS - Access to healthcare

Learning Outcomes
• Explain the concepts of access and equity in health care • Differentiate between accessibility and availability of health care • Identify the groups of people most at risk of poor access and the ways in which this factor affects health equity • Describe the workforce issues contributing to poor access to healthcare in Australia

Lecture 4

Theme: Foundations of Medicine

Session Outline: PATH - Blood composition and haemopoiesis

Learning Outcomes
• Identify the cellular and non-cellular components of blood • Outline the process of haemopoiesis, sites of haemopoiesis and the key requirements for normal cell production • Describe the normal appearance, structure and function of erythrocytes • Describe the normal appearance of leukocytes in the peripheral blood and their function • Name the routine laboratory tests for blood parameters • Use the correct terms for some common abnormalities

Lecture 5

Theme: Foundations of Medicine

Session Outline: PATH - Haemostasis & coagulation

Learning Outcomes
• Identify factors involved in blood coagulation • Outline processes of blood coagulation (the intrinsic, extrinsic, and common pathways) that lead to the formation of stable fibrin clots • Outline the process of fibrinolysis • Name the routine laboratory tests for coagulation studies • Identify some common coagulopathies

Lecture 6

Theme: Foundations of Medicine

Session Outline: PHARM - Anticancer drugs

Learning Outcomes
• Identify general properties of cancer cells • Outline the principles of cancer therapy • Identify targets for anticancer drugs • Describe the mechanism of action of classic cytotoxic drugs • Discuss the major adverse effects of anticancer drug therapy • Discuss strategies for the management of adverse effects • Describe the mechanism of action of novel anticancer drugs • Describe common clinical approaches to drug therapy and the limitations of cancer chemotherapy

Week 9

Lecture 1

Theme: Foundations of Medicine

Session Outline: PATH - Cellular response to injury (Parts I-III)

Learning Outcomes
• Identify the possible outcomes for cells subjected to harmful stimuli or injurious agents • Consider the various effects of stimuli or injury on tissues • Learn and understand the terms; hyperplasia, hypertrophy, atrophy and metaplasia • Define different tissue types; labile, stable, permanent • Identify causes of cell injury

Lecture 2

Theme: Foundations of Medicine

Session Outline: MICRO - Introduction to microbes

Learning Outcomes
• Describe the three domain structure of the living world • Compare the structure and functions of prokaryotes and eukaryotes • Identify the differences between bacteria and viruses • Articulate the different methods that bacteria and viruses use to replicate

Lecture 3

Theme: Foundations of Medicine

Session Outline: MICRO - Natural barriers

Learning Outcomes

• Describe the natural barriers of the human body to infection, and discuss the innate defence mechanisms:
  ○ Anatomical
  ○ Mechanical
  ○ Antimicrobial/Chemical factors
  ○ Cellular components
• Appreciate the role of the normal flora / microbiota of the human body, and where it is present
• Define the different host-parasite interactions which occur in humans

Lecture 4

Theme: Foundations of Medicine

Session Outline: MICRO - Strategies & control

Learning Outcomes
• Explain the different parts in the chain of infection • Describe ways to control microbial growth • Describe the principles and agents used for sterilization disinfection and antisepsis • Relate the importance of the control of nosocomial infections in the healthcare setting with particular reference to hand hygiene

Lecture 5

Theme: Foundations of Medicine

Session Outline: MICRO - Fungal infections

Learning Outcomes
• Appreciate the importance and impact of fungal infections globally, particularly in the immunocompromised • Explain the transmission and spread of fungal infections • Describe common types of fungal infections and their diagnosis • Discuss using examples fungal pathogens of major importance • Appreciate the impacts of fungal infections in the healthcare setting and the community

Week 10

Lecture 1

Theme: Personal and Professional Development

Session Outline: HEP - Environment

Learning Outcomes
• Understand the many ways in which one can look at what environment is • Appreciate the impact of environment on many aspects of health • Consider the ways in which one can create a healthy environment

Lecture 2

Theme: Population, Society & Illness

Session Outline: HKS - Refugee and asylum seeker health

Learning Outcomes
• Outline the health and social issues faced by refugees and asylum seekers in the community • Describe the specific health issues faced by asylum seekers incarcerated in detention centers in Australia • Locate the appropriate clinical resources available in refugee and asylum seeker health in Australia • Demonstrate an introductory understanding of the current political environment regarding the entry of asylum seekers into Australia

Lecture 3

Theme: Foundations of Medicine

Session Outline: IMMUN - Innate responses to pathogens

Learning Outcomes
• Explain the barriers to prevent infection • Identify the cells that comprise the innate immune system • Describe pathogen profiling as a quick and dirty way to differentiate pathogens from self-antigens • Describe how immune cells are directed to sites of infection

Lecture 4

Theme: Foundations of Medicine

Session Outline: IMMUN - The adaptive immune system

Learning Outcomes
• Explain how antigen is processed and presented • Explain how B and T cell receptors identify their specific antigen • Describe the functions of CD4 and CD8 T cells • Describe the function of B cells and identify the different antibodies produced

Lecture 5

Theme: Foundations of Medicine

Session Outline: IMMUN - Infection and immunity

Learning Outcomes
• Compare and contrast the innate and adaptive immune responses • Identify the key immune players and explain their functions during SARS-CoV-2 infection • Describe an allergic immune response to pollen • Explain the mechanism by which cancer cells evade the immune response through checkpoint blockade

Lecture 6

Theme: Foundations of Medicine

Session Outline: HD - Normal ageing

Learning Outcomes
Part 1 (Social Context) • Understand the concept of healthy aging • Understand and apply the Competence-environmental stress model • Understand the Preventative and corrective proactivity model • Understand the key developmental issues relating to normal aging **Part 2 (Personal Context)** • Apply the Lifespan development theory to aging • Apply Bronfenbrenner's ecological approach to aging • Describe the physiological and cognitive changes associated with aging • Understand the contribution of lifestyle and chronic disease to aging • Describe the four Mental health syndromes (Depression, Anxiety, Cognitive impairment and Psychosis) as they present in older adults

Lecture 7

Theme: Foundations of Medicine

Session Outline: HD - Death and dying

Learning Outcomes
• Understand the definition of death • Know the factors influencing the experience of death • Know how to recognize dying - the Surprise Question, the SPICT tool, the illness trajectories • Know the stages of dying • Know care of the dying patient • Understand bereavement and grieving • Understand advanced care planning • Understand voluntary assisted dying • Know care of self

Week 11

Lecture 1

Theme: Personal and Professional Development

Session Outline: HEP - Connectedness

Learning Outcomes
• Understand the relationship between connectedness, social support and health • Understand the potential health advantages and disadvantages dependent upon one's extent and quality of social support • Know some of the mind body mechanisms for understanding the impact of social health on physical health • Know the effects of social support or isolation at different stages of life

Lecture 2

Theme: Personal and Professional Development

Session Outline: HEP - Spirituality

Learning Outcomes
• Define and distinguish between meaning, spirituality and religion • Understand the wide variety of ways in which individuals and cultures explore and express spirituality • Understand the need for tolerance and respect between different individuals and cultural groups • Understand the role of spirituality in healthcare • Know the relevance of these issues in clinical settings

Lecture 3

Theme: Population, Society & Illness

Session Outline: HKS – Health and human rights

Learning Outcomes
• Understand the basic context of human rights • Discuss the relevance of human rights and social justice issues to health • Evaluate the ways in which human rights impact on health

Lecture 4

Theme: Foundations of Medicine

Session Outline: MICRO - Antimicrobial resistance & resistance transfer

Learning Outcomes
• Define antibiotic resistance and appreciate its impact in the health care setting • List the major targets of antimicrobial agents and relate them to the differences between bacteria and eukaryotes • Describe the mechanisms of antibiotic resistance and how it can be transferred • Explain the drivers of antibiotic resistance and how a One-Health approach can combat them

Lecture 5

Theme: Foundations of Medicine

Session Outline: MICRO - Parasites

Learning Outcomes
• Appreciate the wide range of parasitic infections from single celled protozoa to multicellular 10 metre long intestinal worms! • Recognise that the majority of globally important parasitic infections are Neglected Tropical Diseases and are diseases of poor sanitation • Contrast parasites' very complex lifecycles to the simplicity of bacteria and viruses • Describe some examples of protozoan and helminth infections, how they are transmitted and how we diagnose them

Lecture 6

Theme: Foundations of Medicine

Session Outline: PHARM - Introduction to pharmacokinetics

Learning Outcomes
• Describe the basic principles of pharmacokinetics relating to drug absorption, distribution, metabolism and excretion • Explain the advantages and disadvantages of different routes of administration • Discuss how a knowledge of pharmacokinetics of a drug is important when considering the route of administration, the dosing schedule, and ways of achieving therapeutic levels whilst avoiding toxicity • Explain the terms pharmacokinetics, therapeutic index, bioavailability, half life, clearance, plasma protein binding, biotransformation, first order kinetics, zero order kinetics, induction of liver enzymes

Lectures 7 and 8

Theme: Foundations of Medicine

Session Outline: PHARM - Antibiotics and antivirals

Learning Outcomes
Antibacterials: • Describe why selective toxicity is important in anti-infective chemotherapy • Discuss the main principles of selective toxicity and problems associated with the use of antibacterial agents • Discuss the use, mechanisms of action, clinical uses and common adverse effects of examples from each of the major classes of antibacterial drugs
Antivirals: • Discuss the general principles of the treatment of viral infections (including HIV) • Discuss the mechanisms of action, clinical uses and problems associated with the use of specific antiviral drugs

Week 12

Lecture 1

Theme: Population, Society & Illness

Session Outline: HKS - The pharmaceutical industry

Learning Outcomes
• Discuss the access to essential medicines • Demonstrate an understanding of the effects of global pharmaceutical practices on the availability of antiretroviral medications in sub-Saharan African countries • Recognize the marketing practices employed by pharmaceutical companies when dealing with medical graduates and students

Lecture 2

Theme: Foundations of Medicine

Session Outline: HD - Cognitive development

Learning Outcomes
• Describe, discuss, and criticize the process of cognitive developmental change in Piaget's theory • Explain how culture, social interaction, and language influence thought based on Vygotsky's theory • Summarise Fischer's dynamic skill framework and compare it to Piaget and Vygotsky models of cognitive development • Compare the thinking pattern of neuro-typical preschool-age children, elementary school children, adolescents and adults

Lecture 3

Theme: Foundations of Medicine

Session Outline: HD - Introduction to health and human behaviour

Learning Outcomes
• Appraise definitions of health and illness • Identify current and relevant historical concepts of health and illness • Apply the biopsychosocial model of illness • Explain the impact of culture on health and illness